

AI Student Admission Assistant Chatbot

Minor Project Documentation

Artificial Intelligence Domain

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Problem Statement

Students and parents frequently seek information about admissions, courses, fees, scholarships, hostel facilities, placements, and contact details. Answering these repetitive queries manually is time-consuming. This project provides an AI-based Student Admission Assistant that responds instantly using rule-based NLP and pattern matching.

Objectives

- Develop an AI-based chatbot.
- Answer common admission queries.
- Use pattern matching and text preprocessing.
- Provide a user-friendly console interface.
- Reduce response time for common questions.

Tools & Technologies

Python 3.13

VS Code

JSON

Difflib

Regular Expressions (re)

Datetime

Random

String

AI Concepts Used

Rule-based chatbot

Pattern matching

Text preprocessing

Similarity matching using difflib
Random response selection

System Flow

Start → Load intents.json → Display welcome message → Accept user input → Preprocess text → Match with intents → Display response or ask user to rephrase → Save chat history → Repeat until exit.

Algorithm

1. Load intents.
2. Display greeting.
3. Accept user input.
4. Preprocess text.
5. Compare with stored patterns.
6. Return best matching response.
7. Save chat history.
8. Log unknown questions.
9. Exit when user types 'exit'.

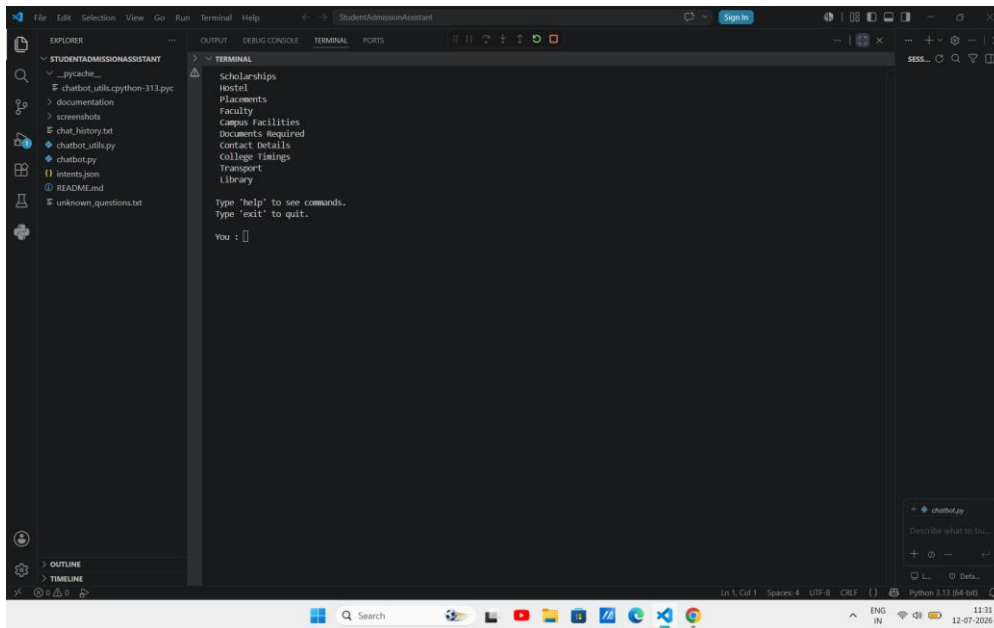
Implementation Steps

1. Created intents dataset.
2. Developed chatbot.py.
3. Implemented preprocessing.
4. Added similarity matching.
5. Added typing animation.
6. Added chat history and unknown question logging.
7. Tested chatbot with multiple queries.

Features

- Greeting
- Admission guidance
- Fees information
- Course information
- Similarity matching
- Chat history
- Unknown question logging
- Session counter
- Exit option

Output Screenshots



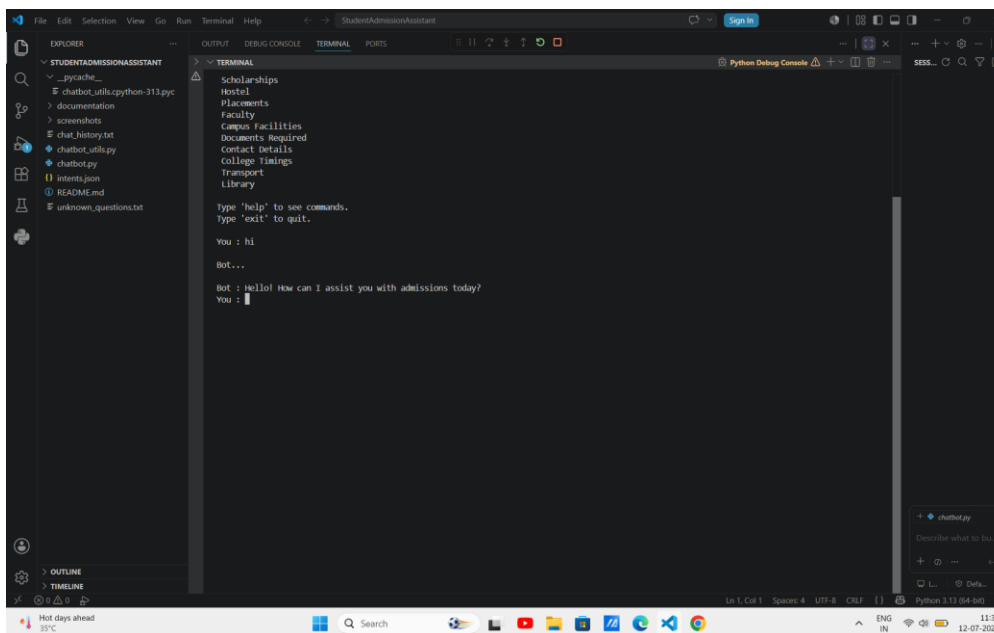
```
File Edit Selection View Go Run Terminal Help StudentAdmissionAssistant
EXPLORER OUTPUT DEBUG CONSOLE TERMINAL PORTS
STUDENTADMISSIONASSISTANT
  __pycache__
  chatbot_utils.cpython-313.pyc
  documentation
  screenshots
  chat_history.txt
  chatbot_utils.py
  chatbot.py
  intents.json
  README.md
  unknown_questions.txt
  OUTLINE
  TIMELINE

TERMINAL
Scholarships
Hostel
Placements
Faculty
Campus Facilities
Documents Required
Contact Details
College Timings
Transport
Library

Type 'help' to see commands.
Type 'exit' to quit.

You : 
```

Screenshot 1 : welcome



```
File Edit Selection View Go Run Terminal Help StudentAdmissionAssistant
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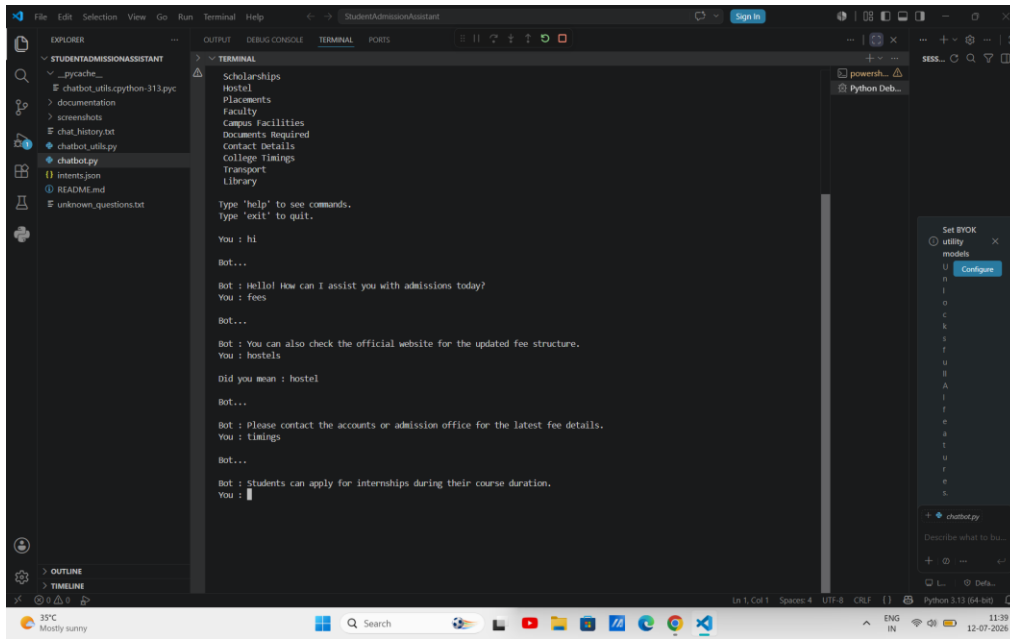
You : hi

Bot...

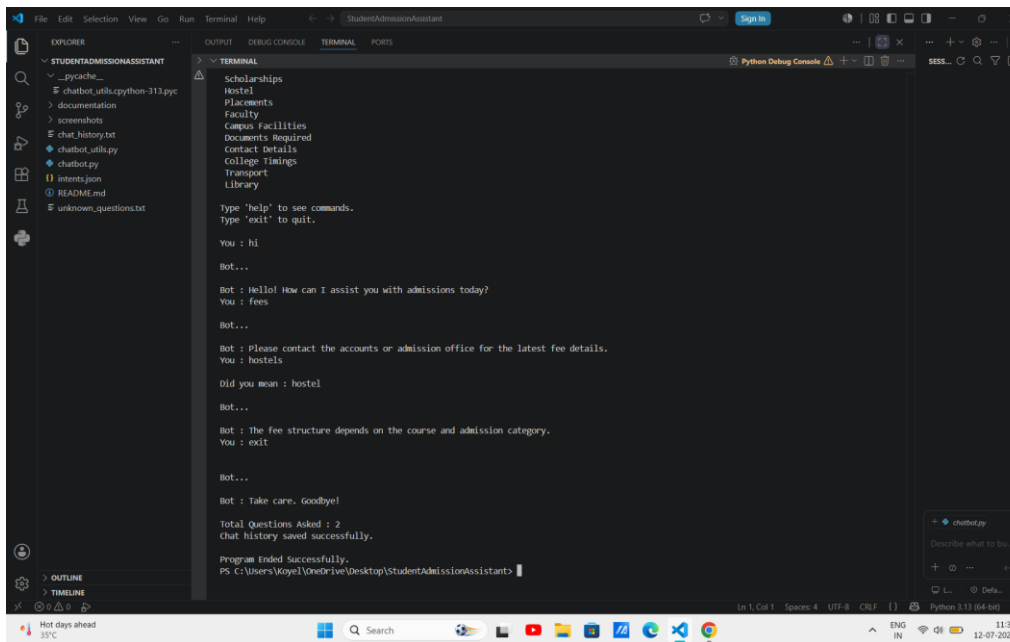
Bot : Hello! How can I assist you with admissions today?

You : 
```

Screenshot 2: Greetings



Screenshots 3: Conversation



Screenshot 4: Exit

Advantages

Fast responses, easy to use, reusable dataset, modular design, easy to extend.

Limitations

Works with predefined intents only and does not connect to a live database.

Future Scope

Web interface, voice support, multilingual chatbot, database integration, machine learning based intent detection.

Conclusion

The AI Student Admission Assistant successfully answers common student admission queries using rule-based NLP techniques. The project demonstrates practical use of Artificial Intelligence concepts in an easy-to-use console application.